SURJECTIVITY PRESERVES POINTS OF FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The criterion for the set of points of finite type through a smooth presentation. Used inside the counted proof (source lines 3750-3753). Expresses the points of finite type of the source and of the target through presentations.
- The corresponding statement for schemes that the image of a point of finite type under a surjective morphism of finite type is again of finite type. Invoked inside the counted proof (source lines 3758-3760) as a cross-chapter pointer. Provides the conclusion at the level of the presenting schemes.

Source reference. Original lemma lemma-base-change-finite-type: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3760–3760. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-finite-type, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-finite-type-points-morphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3751–3751. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-type-points-morphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-identify-finite-type-points: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3757–3757. Navigation pointer only: the counted text refers to the Stacks item lemma-identify-finite-type-points, which belongs to another chapter of the project and is not part

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of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. If f is locally of finite type and surjective, then $f(\mathcal{X}_{ft\text{-}pts}) = \mathcal{Y}_{ft\text{-}pts}$.*

Proof. We have $f(\mathcal{X}_{\text{ft-pts}}) \subset \mathcal{Y}_{\text{ft-pts}}$ by Lemma lemma-finite-type-points-morphism. Let $y \in |\mathcal{Y}|$ be a finite type point. Represent y by a morphism $\mathrm{Spec}(k) \to \mathcal{Y}$ which is locally of finite type. As f is surjective the algebraic stack $\mathcal{X}_k = \mathrm{Spec}(k) \times_{\mathcal{Y}} \mathcal{X}$ is nonempty, therefore has a finite type point $x \in |\mathcal{X}_k|$ by Lemma lemma-identify-finite-type-points. Now $\mathcal{X}_k \to \mathcal{X}$ is a morphism which is locally of finite type as a base change of $\mathrm{Spec}(k) \to \mathcal{Y}$ (Lemma lemma-base-change-finite-type). Hence the image of x in $\mathcal{X}$ is a finite type point by Lemma lemma-finite-type-points-morphism which maps to y by construction. □